

Designing Clinical Research

Feedback Session

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General

- Data analysis.
- Sample size determination.
- LLM use.
 - Developing an eCRF represents a wise use of LLM
- Over-engineered protocols: questionable feasibility
 - Unrealistic large cohorts, intensive follow-up procedures, extensive participant retention activities, extensive psychiatric assessments, multiple diagnostic instruments, whole-exome sequencing, centralized biobanking, multicenter recruitment, amyloid confirmation, score development, validation steps.

Section 1

SI and Lactate

SI and Lactate

- Rationale - target population Mismatch: a protocol relies on evidence from out-of-hospital cardiac arrest patients to justify a study in polytrauma patients.
- No single clearly defined primary objective: mortality prediction, ICU admission, predictor comparison, composite models, correlations, score development.
- Polytrauma definition: ISS, AIS, or the Berlin definition - Subject area expert

SI and Lactate

- Overly restrictive eligibility criteria:
 - Excluding for example elderly, severe head injury, diabetes, substance use reduces generalizability.
 - Excluding patients who die early creates survival bias when mortality is the main outcome.
 - Exclusion of patients with missing data systematically removes the least-investigated patients.
- Restricting enrollment to patients with elevated SI and/or lactate excludes normal-value patients making assessment of predictive value **impossible**.

SI and Lactate

- Primary outcome must be clearly defined: in-hospital mortality, 24-hour mortality, 48-hour mortality, and 30-day mortality - subject-area expert:
- Design inconsistency: prospective, yet data collection is based on date extraction from hospital records.
- Use of inappropriate variables for trauma research: Cardiac arrest variables such as ROSC, bystander CPR, and shockable rhythm are not relevant to trauma populations.
- Major trauma confounders are omitted e.g., GCS, injury pattern, comorbidities, medications, trauma interventions.

SI and Lactate

- Incorrect measurement units: SI and MSI are dimensionless indices, should not be reported using mmol/L.
- Reliance on retrospective hospital records increases risk of information bias and missing data.
- Objectives - design Mismatch: estimating prevalence of polytrauma or shock frequency within a cohort restricted to polytrauma cases.
- Several anticipated findings appear pre-specified, creating concern that conclusions are assumed before data collection.

Section 2

PPD and MoD

PPD and MoD

- Design - Research question Mismatch: XS design to assess “risk”.
- Design descriptions are internally inconsistent: XS design while planning repeated follow-up assessments over several months.
- MoD is inconsistently categorized. Some protocols combine all CS despite evidence that elective and emergency CS may have different associations with PPD.

PPD and MoD

- Selection bias: Recruitment from psychiatric clinics, postnatal clinics, or vaccination clinics may not yield participants representative of the wider postpartum population.
- Excluding women likely to be lost to *follow-up* introduces additional selection bias.
- Use of EPDS-defined controls with exclusion of intermediate scores creates spectrum & selection bias.

PPD and MoD

- Overly restrictive eligibility criteria remove many real-world obstetric cases and limit external validity.
 - The striking example is restriction to uncomplicated PG only.
- Flawed Matching in case-control studies. Matching controls on MoD (the exposure being investigated), will eliminate the association under study.

PPD and MoD

- Inadequate confounders: e.g. marital conflict, intimate partner violence are incompletely addressed
- Internal contradiction: previous depression is often listed as a confounder while women with prior psychiatric illness are excluded from the study.

Section 3

ADHD and MoD

ADHD and MoD

- Cases and controls may not arise from the same source population, creating a major risk of selection bias.
- Insufficient assurance that controls represent the population that produced the cases.
- Recruitment from tertiary psychiatry centers may limit representativeness, since referred ADHD cases may differ from typical community ADHD cases.

ADHD and MoD

- Exposure assessment relies on parental recall years after birth, creating risk of recall bias for perinatal factors.
- Excluding children with parental psychiatric illness may bias results, because parental mental health is itself strongly associated with ADHD risk
- Several protocols include very large numbers of confounders relative to projected sample size, increasing the likelihood of overfitting and unstable regression estimates.

Section 4

Miscellaneous topics

Miscellaneous topics

- Mismatch between study design and research question:
 - Case-control studies to compare effectiveness of smoking cessation interventions.
 - Cross-sectional studies attempting to infer causal relationships or diagnostic delays.
- Objectives must be specific.
- Secondary aims frequently introduce distinct research questions, resulting in objective creep.
- Major internal inconsistencies in study design descriptions: XS, cohort, case-control, or genetic association research in different sections.

Miscellaneous topics

- Outcomes are often poorly defined, subjective, or insufficiently validated.
 - Diagnostic delay and appropriate treatment lack objective verification.
 - Smoking cessation relies primarily on self-report.
 - PPD is measured using screening instruments rather than confirmed diagnoses.

Miscellaneous topics

- Recall bias is a major threat across several protocols.
 - Query recall childbirth experiences, AI use, smoking history, physician consultations, or treatment timelines.
 - Individuals with depression may differentially recall or interpret past experiences.
- Important confounders are inadequately measured or omitted.
 - Pre-pregnancy and antenatal depression in PPD studies.
 - Motivation and nicotine dependence in smoking cessation studies.
 - Clinical and environmental risk factors in genetic studies.

Miscellaneous topics

- Confounding by indication is a major concern in comparative effectiveness studies, because individuals selecting different treatments differ systematically in motivation, disease severity, and baseline characteristics.
- Age at diagnosis is incorrectly used as a proxy for age at disease onset.
- Enthusiasm > Multiple secondary genomic objectives > dilute the primary research question.

Section 5

Future steps
